

The Half Angle Identities and More Double Angle Problems

The Half Angle Identities

I) $\cos(2A) = 1 - 2\sin^2 A$

$$\cos\left[2\left(\frac{A}{2}\right)\right] = 1 - 2\sin^2\left(\frac{A}{2}\right)$$

$$\cos A = 1 - 2\sin^2\left(\frac{A}{2}\right)$$

$$\cos A - 1 = -2\sin^2\left(\frac{A}{2}\right)$$

$$\frac{\cos A - 1}{-2} = \sin^2\left(\frac{A}{2}\right)$$

$$\frac{-\cos A + 1}{2} = \sin^2\left(\frac{A}{2}\right)$$

$$\frac{1 - \cos A}{2} = \sin^2\left(\frac{A}{2}\right)$$

$$\pm \sqrt{\frac{1 - \cos A}{2}} = \sin\left(\frac{A}{2}\right)$$

$$\sin\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 - \cos A}{2}}$$

II) $\cos(2A) = 2\cos^2 A - 1$

$$\cos\left[2\left(\frac{A}{2}\right)\right] = 2\cos^2\left(\frac{A}{2}\right) - 1$$

$$\cos A = 2\cos^2\left(\frac{A}{2}\right) - 1$$

$$\cos A + 1 = 2\cos^2\left(\frac{A}{2}\right)$$

$$\frac{\cos A + 1}{2} = \cos^2\left(\frac{A}{2}\right)$$

$$\frac{1 + \cos A}{2} = \cos^2\left(\frac{A}{2}\right)$$

$$\pm \sqrt{\frac{1 + \cos A}{2}} = \cos\left(\frac{A}{2}\right)$$

$$\cos\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 + \cos A}{2}}$$

- ① If θ is in standard position, and it terminates in QIV ($270^\circ < \theta < 360^\circ$), find the value of each expression below if $\cos \theta = 4/5$.

$$i) \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1+\cos\theta}{2}} = -\sqrt{\frac{1+4/5}{2}} = -\sqrt{\frac{9}{10}}$$

$$\frac{270^\circ}{2} < \frac{\theta}{2} < \frac{360^\circ}{2}$$

$$135^\circ < \frac{\theta}{2} < 180^\circ$$

$$270^\circ < \theta < 360^\circ$$

$$= -\sqrt{\frac{9}{10}}$$

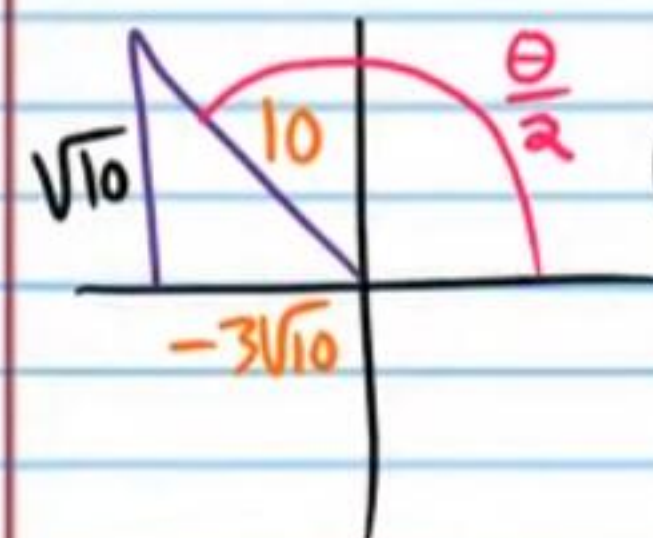
$$= -\frac{\sqrt{9}}{\sqrt{10}}$$

$$= -\frac{3}{\sqrt{10}}$$

$$= \boxed{-\frac{3\sqrt{10}}{10}}$$

$$ii) \sin\left(\frac{\theta}{2}\right) = \boxed{\frac{\sqrt{10}}{10}}$$

$$a^2 + b^2 = c^2$$



$$(-3\sqrt{10})^2 + b^2 = 10^2$$

$$90 + b^2 = 100$$

$$b = \pm\sqrt{10}$$

$$b = \sqrt{10}$$

$$iii) \tan\left(\frac{\theta}{2}\right) = \frac{\sqrt{10}}{-3\sqrt{10}} = \boxed{-\frac{1}{3}}$$

② If B is in standard position, and it terminates in QII ($90^\circ < B < 180^\circ$), find the value of each expression below if $\cos B = -5/6$.

$$i) \sin\left(\frac{B}{2}\right) = \pm \sqrt{\frac{1 - \cos B}{2}} = \sqrt{\frac{1 - \cos B}{2}} = \sqrt{\frac{1 - (-5/6)}{2}}$$

$$\begin{array}{|l} 90^\circ < B < 180^\circ \\ \hline \frac{90^\circ}{2} < \frac{B}{2} < \frac{180^\circ}{2} \\ 45^\circ < \frac{B}{2} < 90^\circ \end{array}$$

$$= \sqrt{\frac{1 + 5/6}{2}}$$

$$= \sqrt{\frac{11}{6}}$$

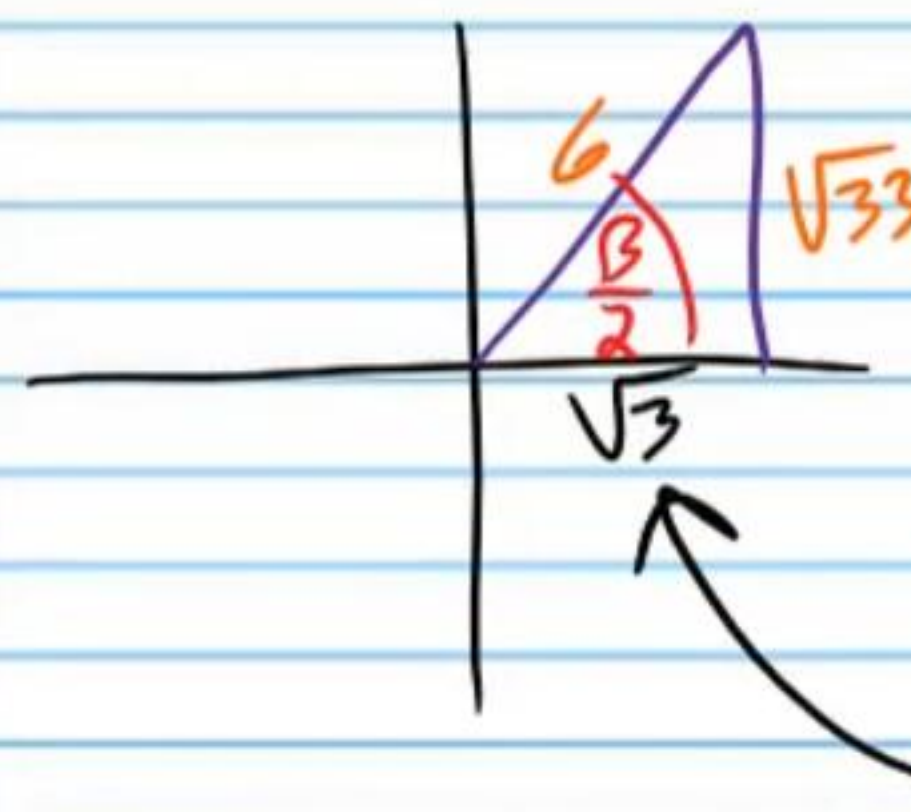
$$= \sqrt{\frac{11}{12}}$$

$$= \frac{\sqrt{11}}{\sqrt{12}}$$

$$= \frac{\sqrt{11}}{2\sqrt{3}}$$

$$= \boxed{\frac{\sqrt{33}}{6}}$$

$$ii) \cos\left(\frac{B}{2}\right) = \boxed{\frac{\sqrt{3}}{6}}$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ (\sqrt{33})^2 + b^2 &= 6^2 \\ b &= \pm\sqrt{3} \\ b &= \sqrt{3} \end{aligned}$$

$$iii) \csc\left(\frac{B}{2}\right) = \frac{6}{\sqrt{33}} = \frac{6\sqrt{33}}{33} = \boxed{\frac{2\sqrt{33}}{11}}$$

- ③ If α is in standard position, and it terminates in QIII ($180^\circ < \alpha < 270^\circ$), find the value of each expression below if $\cos \alpha = -\frac{1}{3}$.

$$i) \cos\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1+\cos\alpha}{2}} = -\sqrt{\frac{1+(-\frac{1}{3})}{2}} = -\sqrt{\frac{\frac{2}{3}}{2}}$$

$$\frac{180^\circ}{2} < \frac{\alpha}{2} < \frac{270^\circ}{2}$$

$$90^\circ < \frac{\alpha}{2} < 135^\circ$$

$$180^\circ < \alpha < 270^\circ$$

$$= -\sqrt{\frac{2}{6}}$$

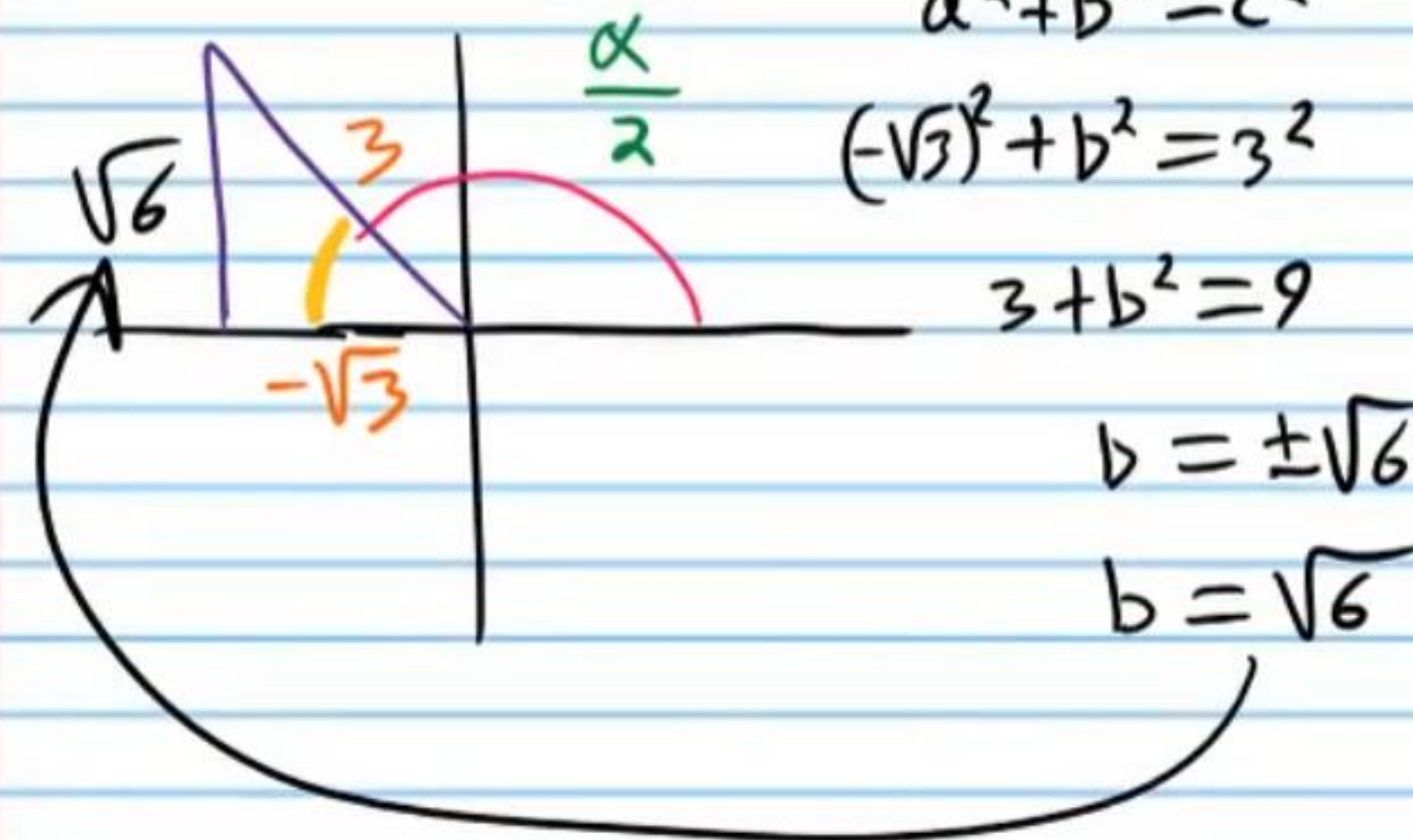
$$= -\sqrt{\frac{1}{3}}$$

$$= -\frac{\sqrt{1}}{\sqrt{3}}$$

$$= -\frac{1}{\sqrt{3}}$$

$$= \boxed{-\frac{\sqrt{3}}{3}}$$

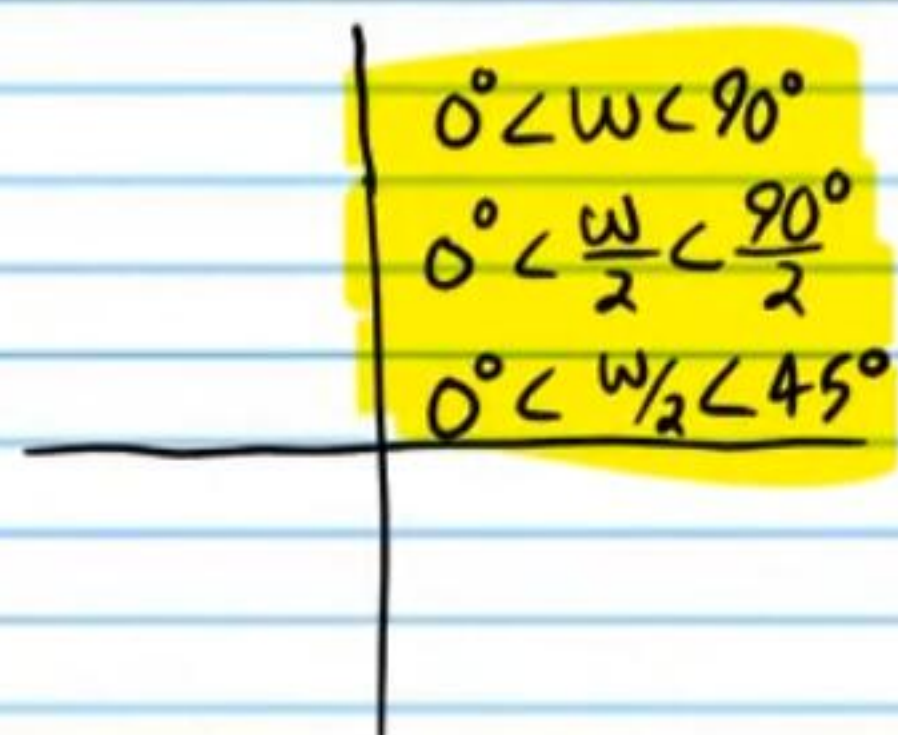
$$ii) \sin\left(\frac{\alpha}{2}\right) = \boxed{\frac{\sqrt{6}}{3}}$$



$$iii) \sec\left(\frac{\alpha}{2}\right) = \frac{3}{-\sqrt{3}} = -\frac{3\sqrt{3}}{3} = \boxed{-\sqrt{3}}$$

④ If W is in standard position, and it terminates in QI ($0^\circ < W < 90^\circ$), find the value of each expression below if $\cos W = \frac{2}{3}$

$$i) \sin\left(\frac{W}{2}\right) = \pm \sqrt{\frac{1 - \cos W}{2}} = \sqrt{\frac{1 - \frac{2}{3}}{2}} = \sqrt{\frac{\frac{1}{3}}{2}}$$



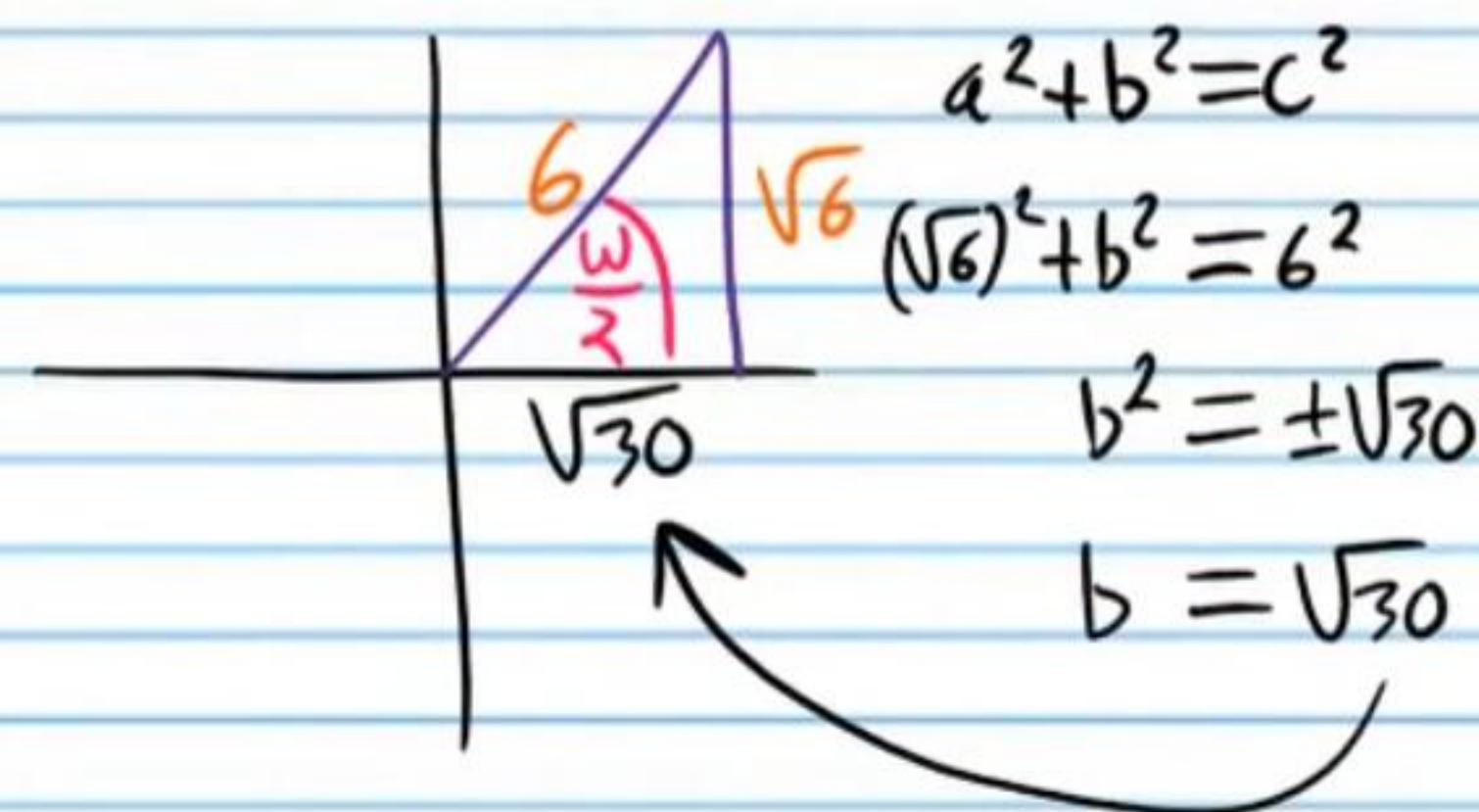
$$= \sqrt{\frac{1}{6}}$$

$$= \frac{\sqrt{1}}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}}$$

$$= \boxed{\frac{\sqrt{6}}{6}}$$

$$ii) \cos\left(\frac{W}{2}\right) = \boxed{\frac{\sqrt{30}}{6}}$$



$$iii) \cot\left(\frac{W}{2}\right) = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{\frac{30}{6}} = \boxed{\sqrt{5}}$$

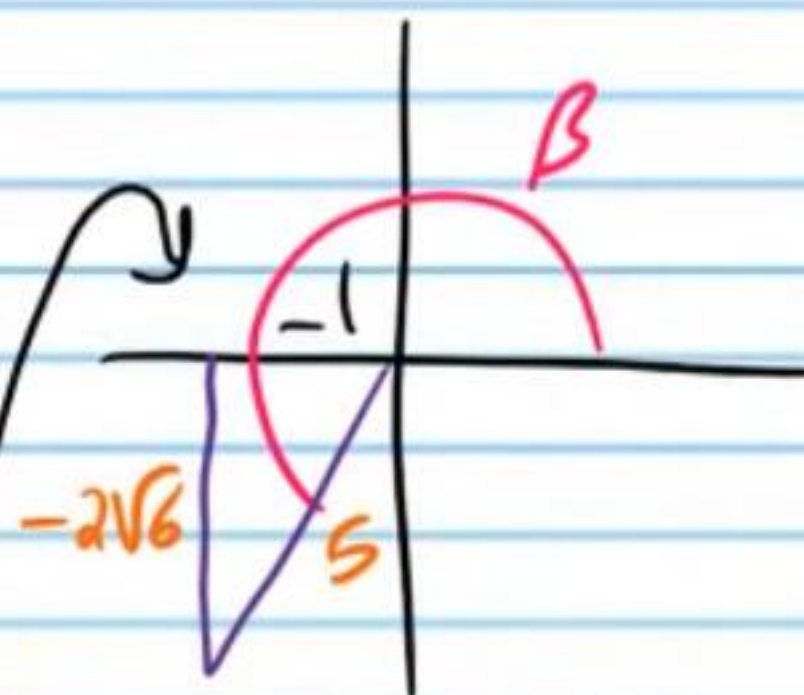
- ⑤ If β is in standard position, and it terminates in QIII ($180^\circ < \beta < 270^\circ$), find the value of each expression below if $\sin \beta = -\frac{2\sqrt{6}}{5}$.

$$i) \sin\left(\frac{\beta}{2}\right) = \pm \sqrt{\frac{1 - \cos \beta}{2}} = \sqrt{\frac{1 - (-\frac{1}{5})}{2}} = \sqrt{\frac{\frac{6}{5}}{2}}$$

$$\frac{180^\circ}{2} < \frac{\beta}{2} < \frac{270^\circ}{2}$$

$$90^\circ < \frac{\beta}{2} < 135^\circ$$

$$180^\circ < \beta < 270^\circ$$



$$= \sqrt{\frac{6}{10}}$$

$$= \sqrt{\frac{3}{5}}$$

$$= \frac{\sqrt{3}}{\sqrt{5}}$$

$$= \boxed{\frac{\sqrt{15}}{5}}$$

$$a^2 + b^2 = c^2$$

$$(-2\sqrt{6})^2 + b^2 = 5^2$$

$$24 + b^2 = 25$$

$$b^2 = 1$$

$$b = \pm 1$$

$$b = -1$$

$$(-2\sqrt{6})(-2\sqrt{6}) = 4\sqrt{36}$$

$$= 4 \cdot 6$$

$$= 24$$

$$ii) \cos\left(\frac{\beta}{2}\right) = \boxed{\frac{-\sqrt{10}}{5}}$$

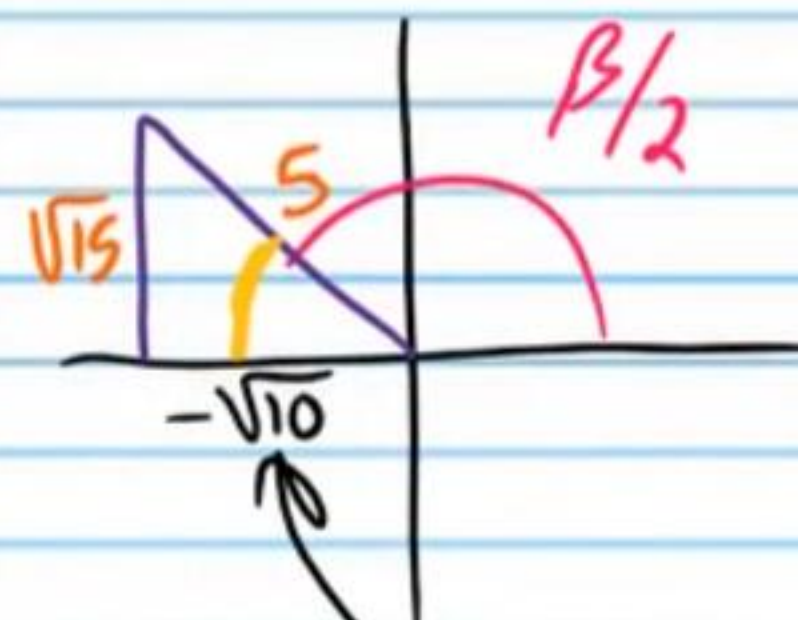
$$a^2 + b^2 = c^2$$

$$(\sqrt{15})^2 + b^2 = 5^2$$

$$15 + b^2 = 25$$

$$b = \pm \sqrt{10}$$

$$b = -\sqrt{10}$$



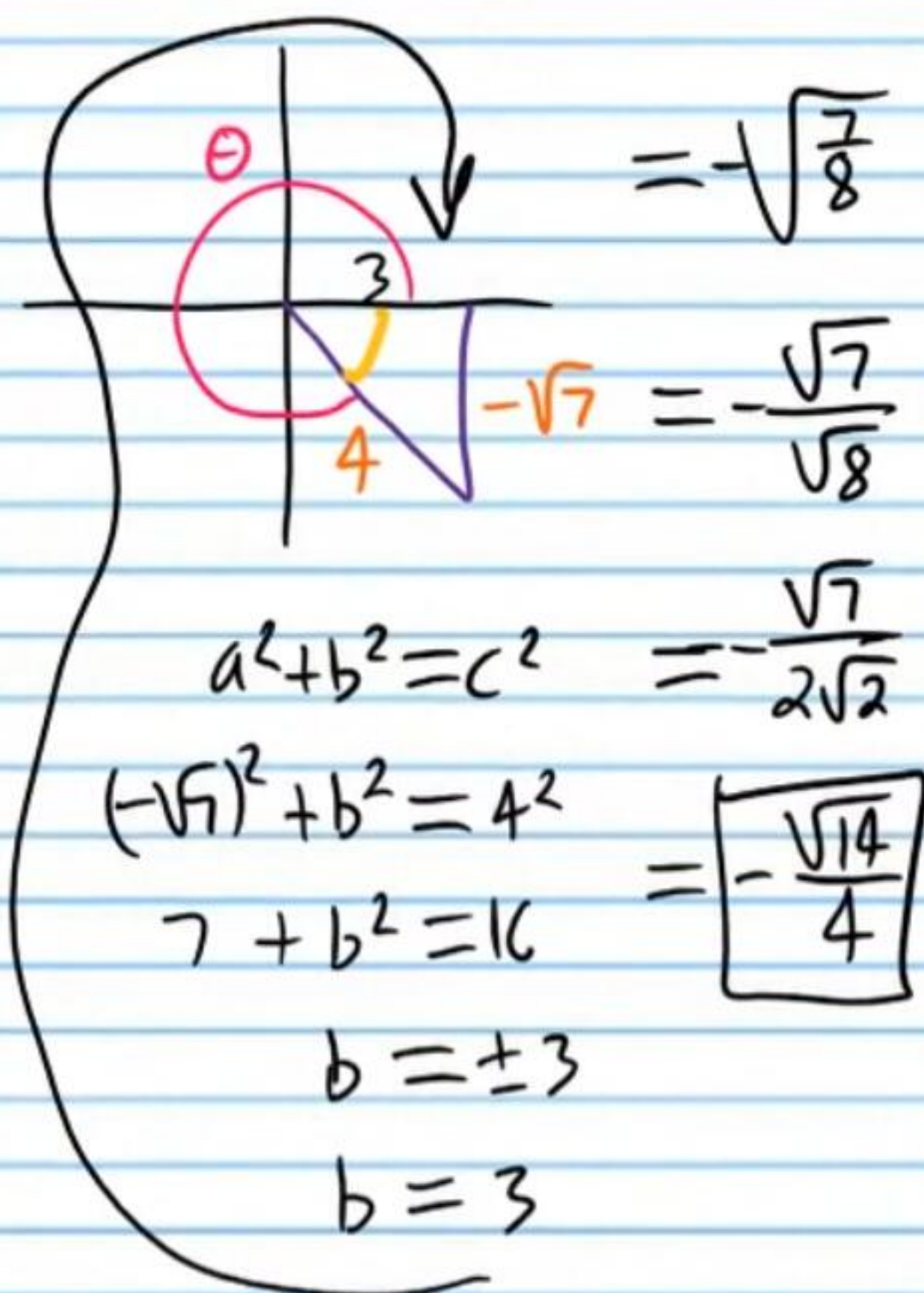
- ⑥ If θ is in standard position, and it terminates in QIV ($270^\circ < \theta < 360^\circ$), find the value of each expression below if $\sin \theta = -\frac{\sqrt{7}}{4}$.

$$i) \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1+\cos\theta}{2}} = -\sqrt{\frac{1+\frac{3}{4}}{2}} = -\sqrt{\frac{\frac{7}{4}}{2}}$$

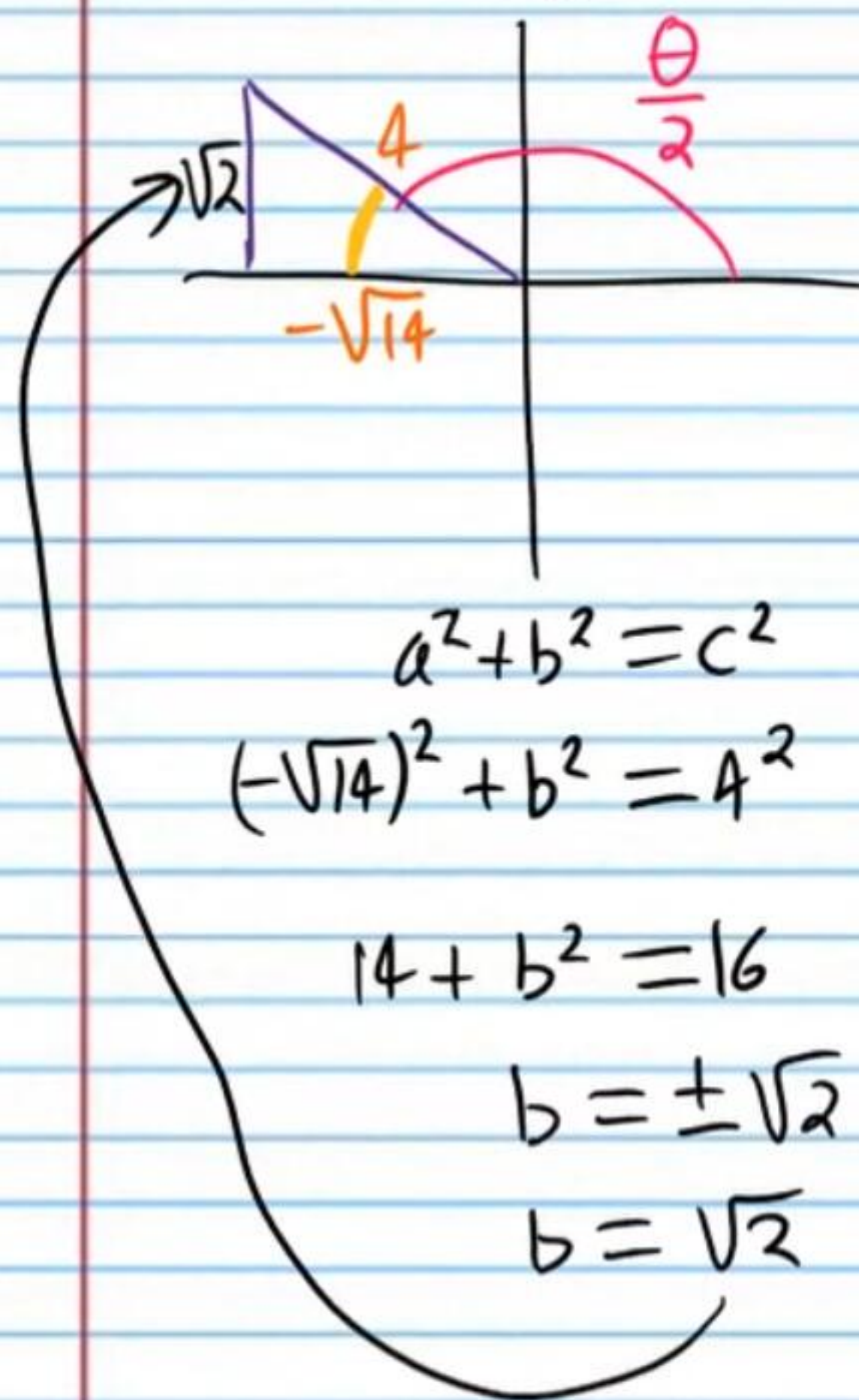
$$\frac{270^\circ}{2} < \frac{\theta}{2} < \frac{360^\circ}{2}$$

$$135^\circ < \frac{\theta}{2} < 180^\circ$$

$$270^\circ < \theta < 360^\circ$$



$$ii) \sin\left(\frac{\theta}{2}\right) = \boxed{\frac{\sqrt{2}}{4}}$$



More Double Angle Problems

$$\sin(2A) = 2\sin A \cos A \quad \cos(2A) = \cos^2 A - \sin^2 A$$

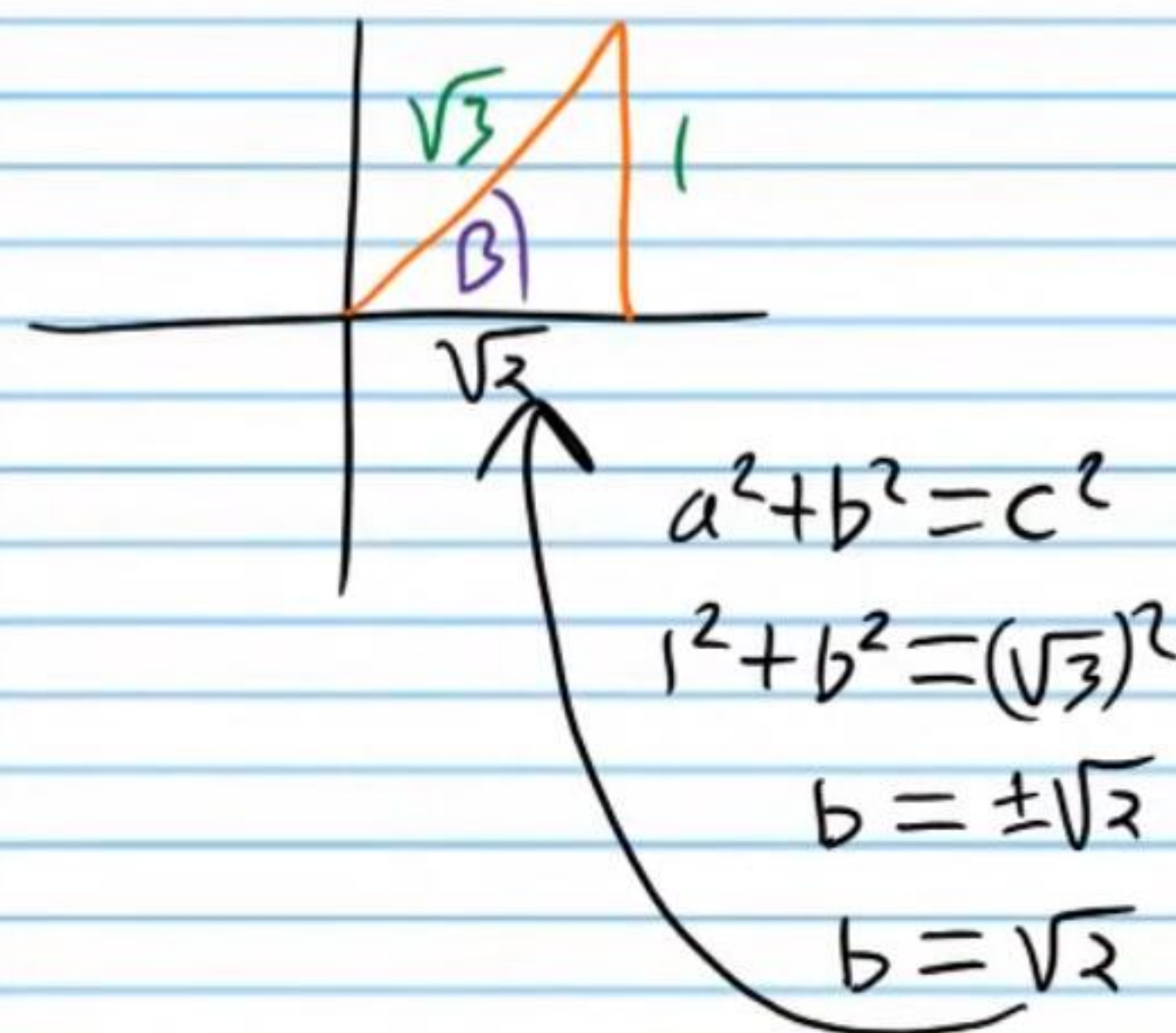
$$\cos(2A) = 1 - 2\sin^2 A \quad \cos(2A) = 2\cos^2 A - 1$$

- ⑦ If B is in standard position, and it terminates in QI ($0^\circ < B < 90^\circ$), find the value of each expression below if $\sin B = \frac{1}{\sqrt{3}}$.

$$\begin{aligned} \text{i) } \cos(2B) &= 1 - 2\sin^2 B = 1 - 2\left(\frac{1}{\sqrt{3}}\right)^2 = 1 - 2\left(\frac{1}{3}\right) \\ &= 1 - \frac{2}{3} \\ &= \boxed{\frac{1}{3}} \end{aligned}$$

- ii) Find the value of $\sin(2B)$ using the double angle formula below.

$$\sin(2B) = 2\sin B \cos B = 2\left(\frac{1}{\sqrt{3}}\right)\left(\frac{\sqrt{2}}{\sqrt{3}}\right) = \boxed{\frac{2\sqrt{2}}{3}}$$



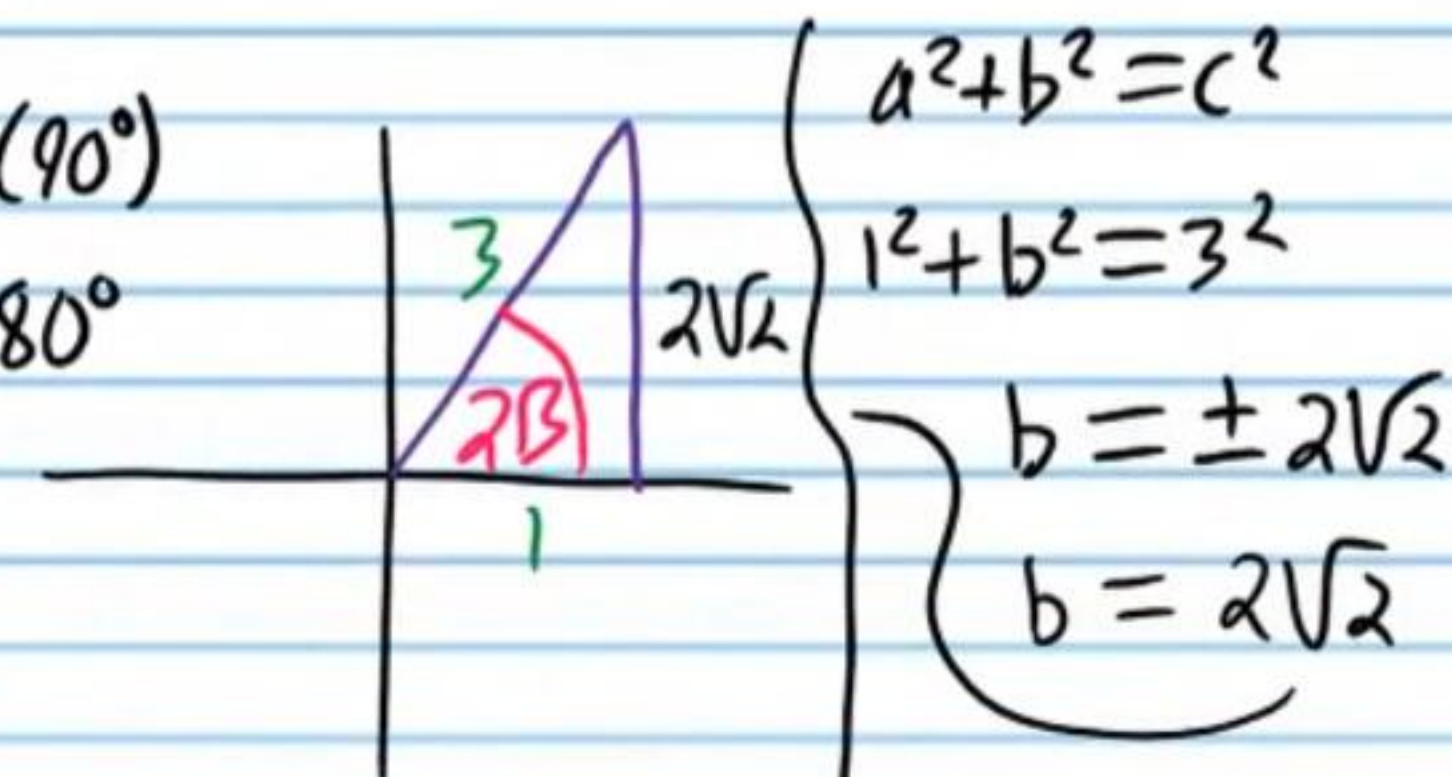
- iii) Find the value of $\sin(2B)$ using the value of $\cos(2B)$ that you found in part one of this problem.

$$\sin(2B) = \boxed{\frac{2\sqrt{2}}{3}}$$

$$0^\circ < B < 90^\circ$$

$$2(0^\circ) < 2B < 2(90^\circ)$$

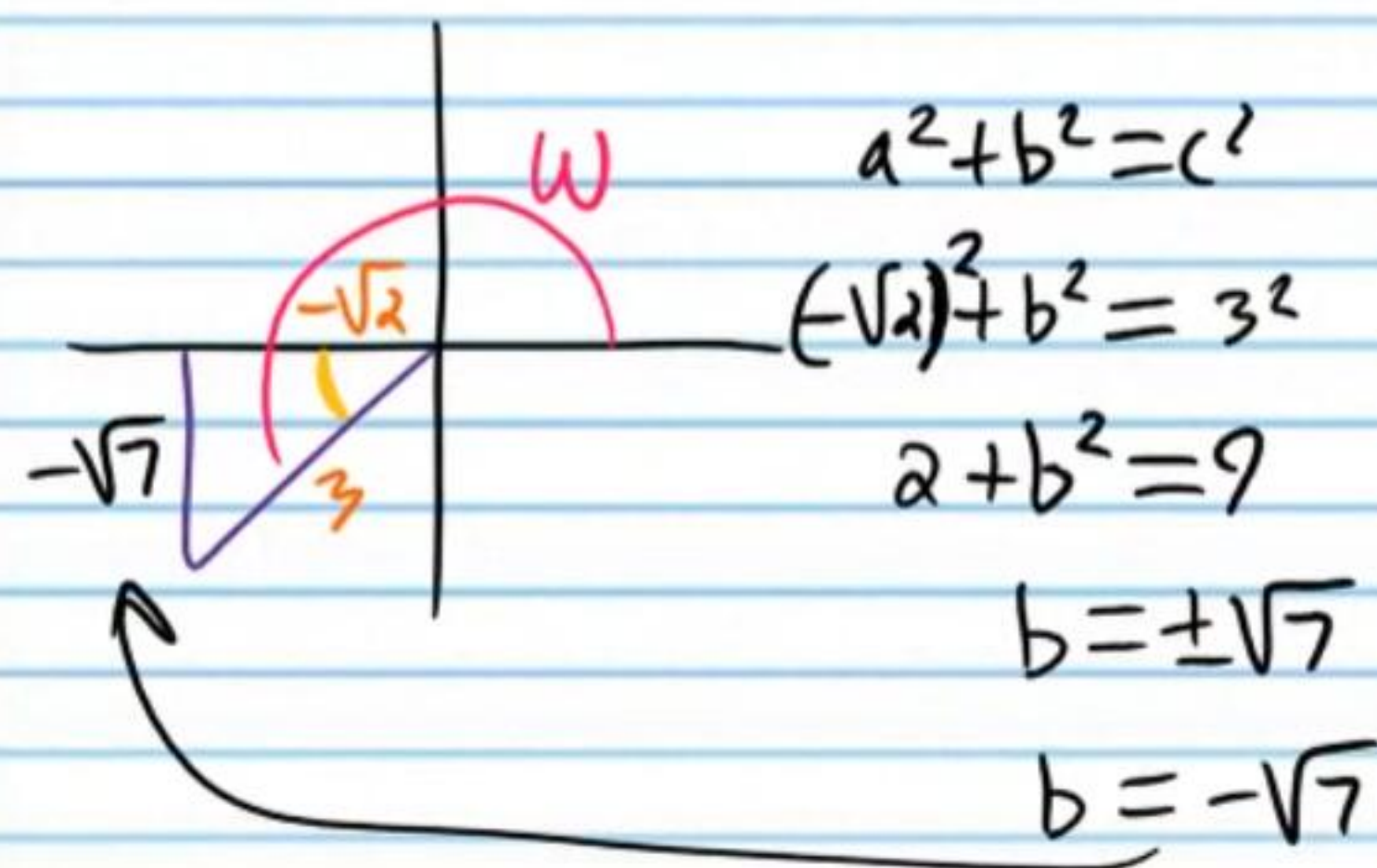
$$0^\circ < 2B < 180^\circ$$



⑧ If W is in standard position, and it terminates in QIII ($180^\circ < W < 270^\circ$), find the value of each expression below if $\cos W = -\frac{\sqrt{2}}{3}$.

$$\begin{aligned} \text{i) } \cos(2W) &= 2\cos^2 W - 1 = 2\left(-\frac{\sqrt{2}}{3}\right)^2 - 1 = 2\left(\frac{2}{9}\right) - 1 \\ &= \frac{4}{9} - 1 \\ &= \boxed{-\frac{5}{9}} \end{aligned}$$

ii) Find the value of $\sin(2W)$ using the double angle formula below.

$$\sin(2W) = 2\sin W \cos W = 2\left(-\frac{\sqrt{7}}{3}\right)\left(-\frac{\sqrt{2}}{3}\right) = \boxed{\frac{2\sqrt{14}}{9}}$$


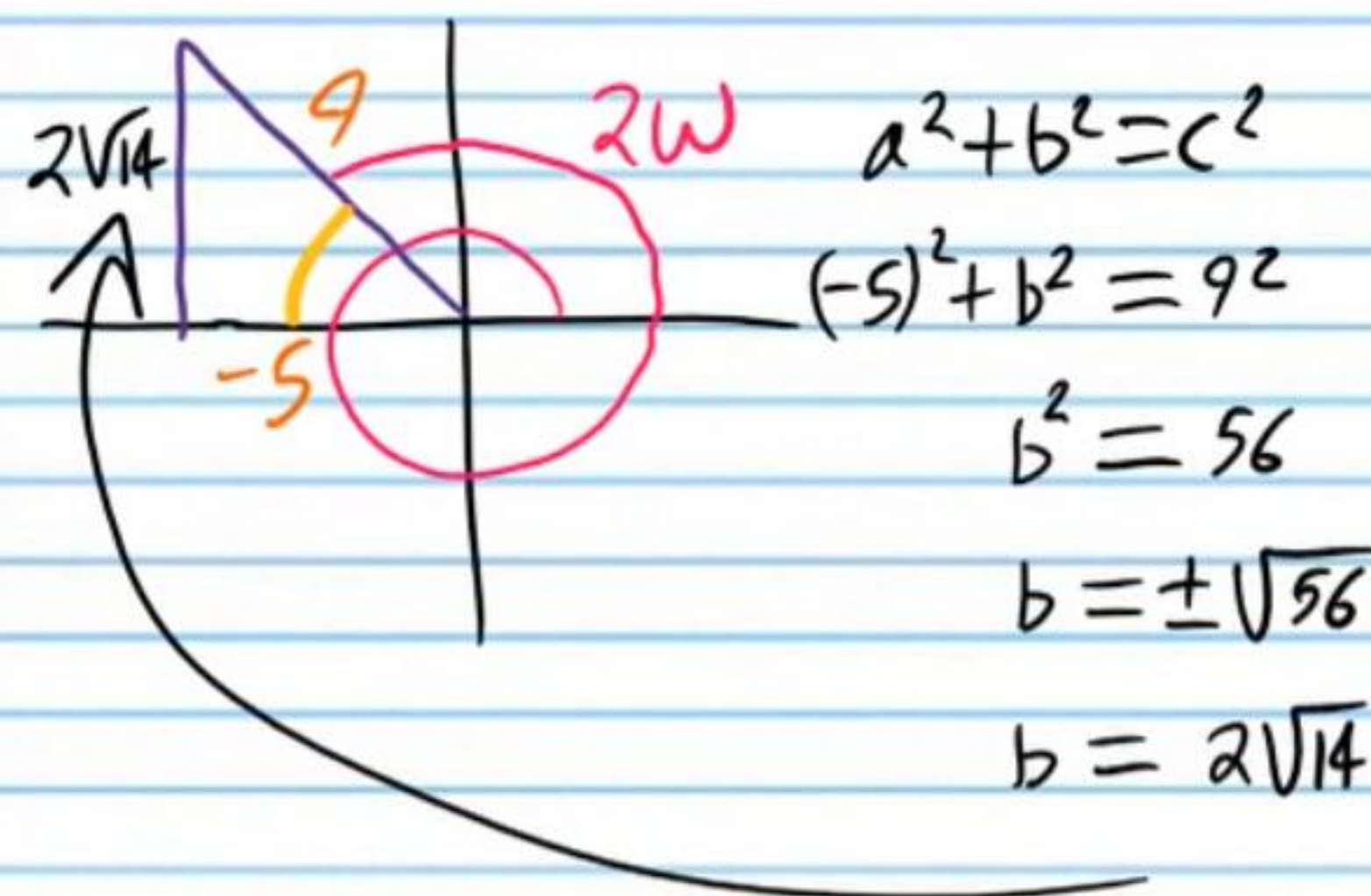
iii) Find the value of $\sin(2W)$ using the value of $\cos(2W)$ that you found in part one of this problem.

$$180^\circ < W < 270^\circ$$

$$\sin(2W) = \boxed{\frac{2\sqrt{14}}{9}}$$

$$2(180^\circ) < 2W < 2(270^\circ)$$

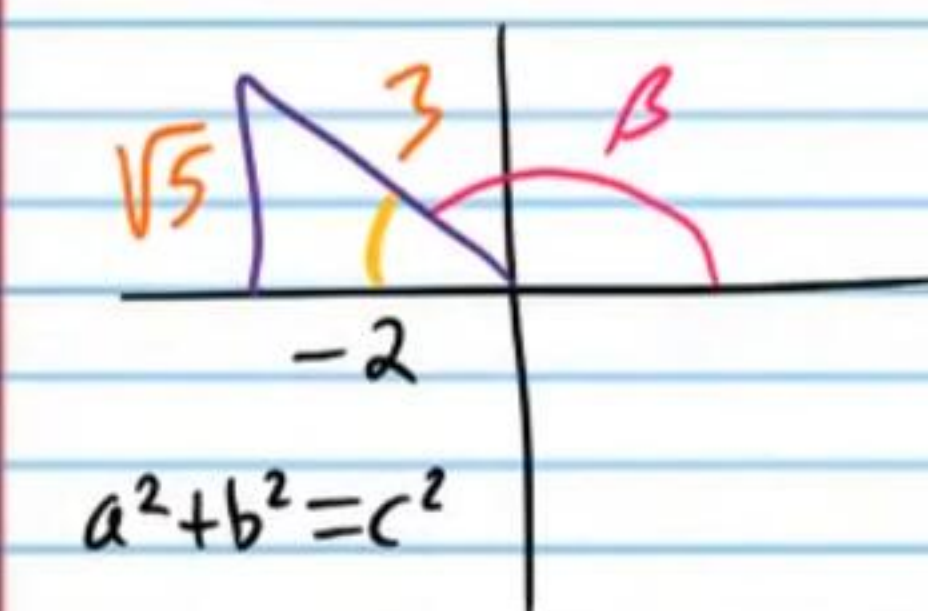
$$360^\circ < 2W < 540^\circ$$



- ⑨ If β is in standard position, and it terminates in QII ($90^\circ < \beta < 180^\circ$), find the value of each expression below if $\sin \beta = \frac{\sqrt{5}}{3}$.

$$\begin{aligned} \text{i) } \cos(2\beta) &= 1 - 2\sin^2 \beta = 1 - 2\left(\frac{\sqrt{5}}{3}\right)^2 = 1 - 2\left(\frac{5}{9}\right) \\ &= 1 - \frac{10}{9} \\ &= \boxed{-\frac{1}{9}} \end{aligned}$$

- ii) Find the value of $\sin(2\beta)$ using the double angle formula below.
- $$\sin(2\beta) = 2\sin\beta\cos\beta = 2\left(\frac{\sqrt{5}}{3}\right)\left(-\frac{2}{3}\right) = \boxed{-\frac{4\sqrt{5}}{9}}$$



$$a^2 + b^2 = c^2$$

$$(\sqrt{5})^2 + b^2 = 3^2$$

$$b = \pm 2$$

$$b = -2$$

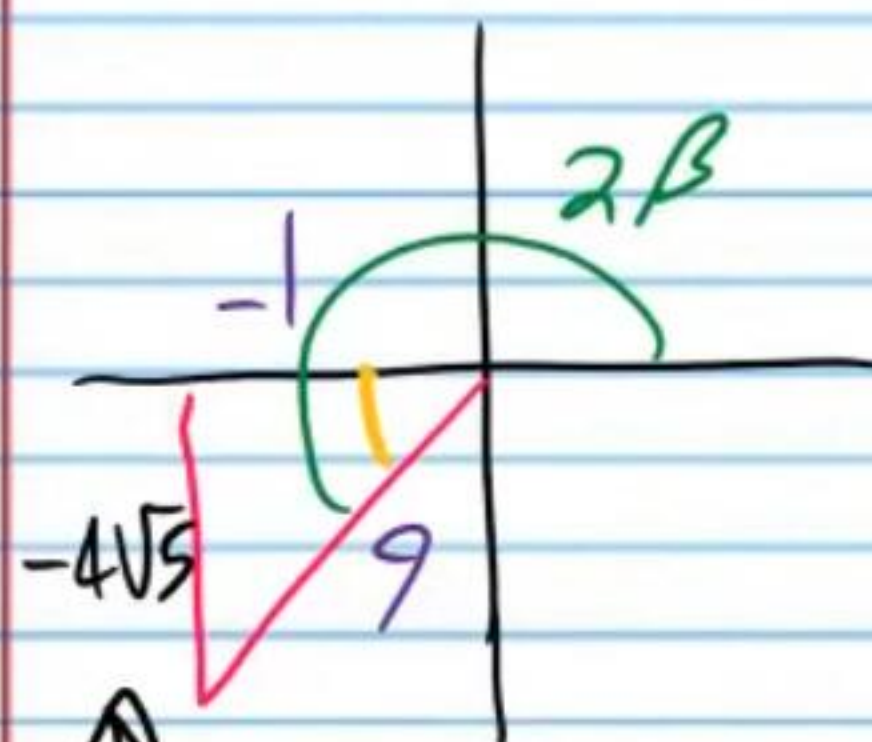
- iii) Find the value of $\sin(2\beta)$ using the value of $\cos(2\beta)$ that you found in part one of this problem.

$$90^\circ < \beta < 180^\circ$$

$$2(90^\circ) < 2\beta < 2(180^\circ)$$

$$180^\circ < 2\beta < 360^\circ$$

$$\sin(2\beta) = \boxed{-\frac{4\sqrt{5}}{9}}$$



$$a^2 + b^2 = c^2$$

$$(-1)^2 + b^2 = 9^2$$

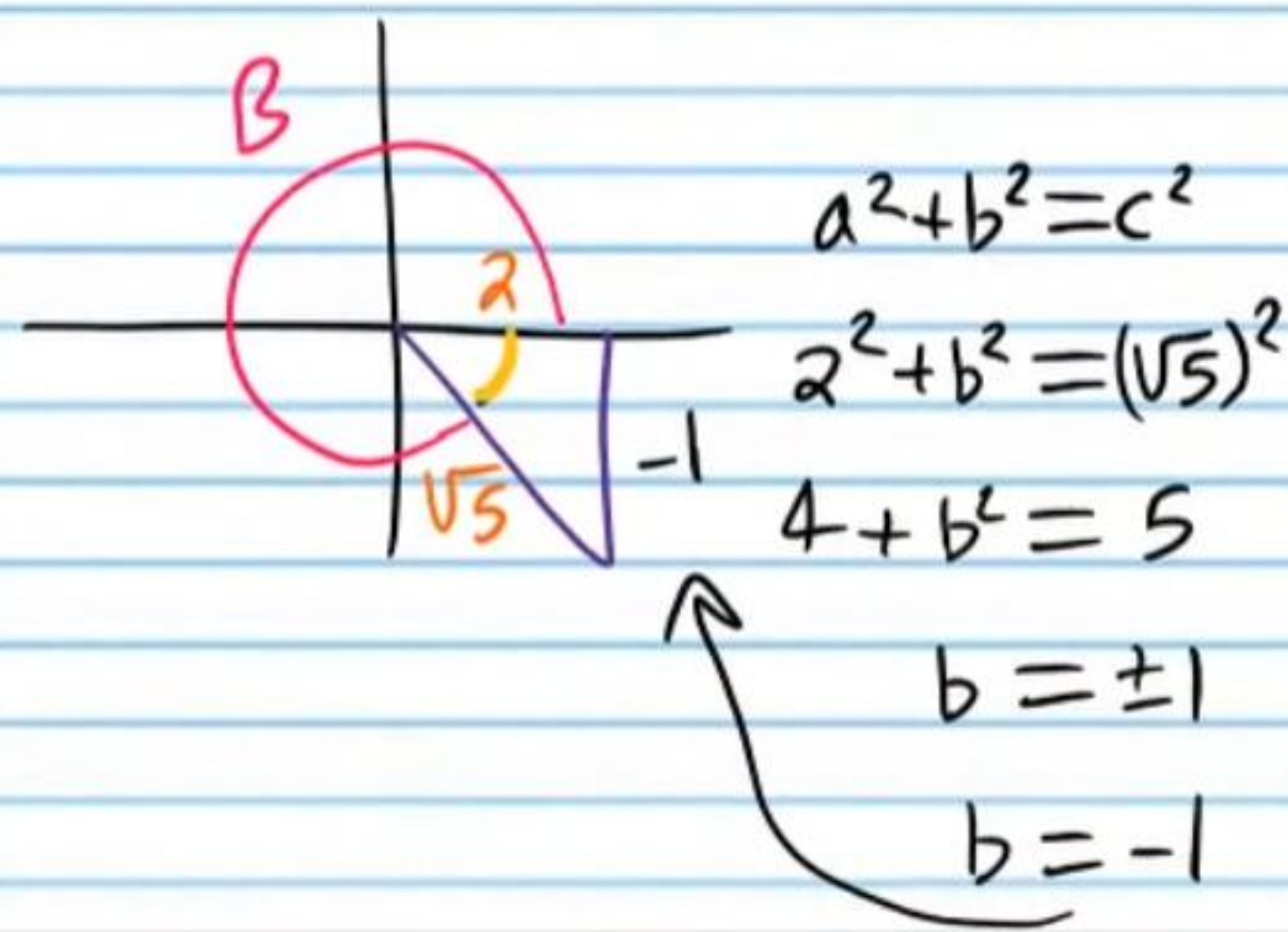
$$b = \pm\sqrt{80}$$

$$b = -4\sqrt{5}$$

⑩ If B is in standard position, and it terminates in QIV ($270^\circ < B < 360^\circ$), find the value of each expression below if $\cos B = \frac{2}{\sqrt{5}}$.

$$\begin{aligned} \text{i) } \cos(2B) &= 2\cos^2 B - 1 = 2\left(\frac{2}{\sqrt{5}}\right)^2 - 1 = 2\left(\frac{4}{5}\right) - 1 \\ &= \frac{8}{5} - 1 \\ &= \boxed{\frac{3}{5}} \end{aligned}$$

ii) Find the value of $\sin(2B)$ using the double angle formula below.

$$\sin(2B) = 2\sin B \cos B = 2\left(\frac{1}{\sqrt{5}}\right)\left(\frac{2}{\sqrt{5}}\right) = \boxed{-\frac{4}{5}}$$


iii) Find the value of $\sin(2B)$ using the value of $\cos(2B)$ that you found in part one of this problem.

$$270^\circ < B < 360^\circ$$

$$2(270^\circ) < 2B < 2(360^\circ)$$

$$540^\circ < 2B < 720^\circ$$

$$\sin(2B) = \boxed{-\frac{4}{5}}$$

